

Professional Practices

Lecture 5

Finance & Accounting

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Outline

Financial Management

- Introduction
- Need of Capital
- Sources of Funds
- Budgeting and Monitoring
- Profit
- Revenue
- Revenue VS Profit
- Financial Image

Introduction

- ❑ However good the quality of its products or services, no organization can be successful for any length of time unless its finances are soundly managed.
- ❑ Many young software engineers are attracted by the idea of starting their own company.

Need of Capital

- ❑ A group of new or recent graduates in computing decide to set up their own company to provide software services and their intention is typically to offer contract hire services
- ❑ A client is unlikely to pay an invoice within less than one month of receiving it. Some large companies are notorious for not paying invoices for as much as six or even twelve months.
- ❑ There will be a need to have some money with which to start the venture



Need of Capital (continued)

- ❑ The group needs enough cash in hand to be able to live for at least three months. Additional money will be needed for the expenses of starting the company
- ❑ For large projects or packages, a much larger sum of money is likely to be needed while they are being developed because there will be no revenue coming into the company.

Need of Capital (continued)

- ❑ For starting period cash will be needed for:
 - Salaries
 - rent rates, heating and lighting of the premises used
 - equipment and consumables
 - costs of advertising and marketing the products
 - miscellaneous expenses, ranging from company stationery to travelling expenses



Need of Capital (continued)

❑ How does one set about raising this money? The first step is to produce a ***business plan***.

❑ It typically contains:

- a description of what the company will be doing, together with information to show that it is technically feasible and that founders of the company have the necessary expertise

- an assessment of the size of the market and the competition

- a prediction of the financial performance of the company



Sources of funds

□ They can be grouped into:

- Grants
- Loans
- Sale of Equity

Grants

- ❑ A *grant* is a sum of money given to the company; while the company is obliged to demonstrate that it has been used for the purposes for which it was intended, it is not intended that the grant should ever be paid back to the organization which gave it
- ❑ The availability of grants and other help for new companies depends very much on where the company is located, how many people it expects to employ, and on government policy at the time.



Loans

- ❑ A loan is a sum of money lent to the company; interest is payable on it, at a rate that may be fixed or variable, and the loan is usually for a fixed period
- ❑ The company is liable to pay back the loan and, if the company goes into liquidation, the lender is entitled to recover the loan from the sale of the assets of the company.
- ❑ In most cases, security is required for the loan

Liquidation is the formal process of closing a business, where its assets are sold (liquidated) to pay off debts, creditors, and liabilities. Once all assets are distributed, the company is dissolved and ceases to exist.

Sale of Equity

- ❑ *Equity capital* is money paid to the company in exchange for a share in the ownership of the company
- ❑ Shareholders are at a much greater risk of getting a poor return on their capital or even losing it completely than are lenders but, in compensation for this, they stand to make a greater profit than lenders if all goes well

Budgeting & Monitoring

- ❑ A budget is a prediction of the future financial position of an organization covering , usually, the current or the next financial year
- ❑ The ordinary manager in a company is, however, much more concerned with budgeting for income and expenditure
- ❑ Budgeting is an iterative process

Budgeting & Monitoring

- ❑ The first version of the budget is likely to show expenditure exceeding income, since the operating managers will want to expand their operations while the sales and marketing department will not wish to give hostages to fortune by being over-optimistic about the volume of sales it can generate. Adjustments will have to be made repeatedly until a situation is reached in which budgeted sales exceed budgeted expenditure with a reasonable profit margin; the operational managers are happy that they can service the predicted volume of sales with the budgeted staff levels; and the salesmen are confident that they can produce the predicted sales

Revenue

- Revenue is the total income a business generates from the sale of goods or services related to its primary operations, often called gross income or turnover. It is the top-line figure on an income statement, calculated as price multiplied by quantity sold, before any expenses are deducted.

Key Aspects of Revenue

- **Gross vs. Net:** Gross revenue is total sales, while net revenue deducts items like returns, allowances, and discounts.
- **Formula:** $\text{Revenue} = \text{Price of Goods} \times \text{Quantity Sold}$.
- **Timeframe:** Measured over specific periods (e.g., monthly, quarterly, annually).
- **Distinction from Profit:** High revenue does not guarantee profit; if expenses exceed revenue, a net loss occurs.
- **Types:** In addition to sales, revenue can come from interest, royalties, or fees.

Profit

- Profit is the financial gain earned when the revenue generated from a business activity exceeds the expenses, costs, and taxes involved in sustaining that activity. It serves as a primary measure of a company's success and its ability to grow or provide returns to investors.

Key Types of Profit

Businesses typically track three main levels of profit on an [income statement](#): 

- **Gross Profit:** The remaining revenue after subtracting the **Cost of Goods Sold (COGS)**. It measures how efficiently a company produces its products.
- **Operating Profit:** Also known as **EBIT** (Earnings Before Interest and Taxes), this is gross profit minus all operating expenses like rent, utilities, and marketing.
- **Net Profit:** Often called the "**bottom line**," this is the final amount left after all expenses, including interest and taxes, have been paid.  Investopedia +5

Profit

Profit = Total Revenue - Total Expenses  Coursera +1

Recent Market Context (February 2026)

Recent financial reports from major entities highlight varying profit trends:

- **HBL**: Reported a pre-tax profit of **Rs 148.1 billion** for 2025, a 23% year-over-year increase.
- [Hub Power Company \(HUBC\)](#): Posted a net profit of **Rs 14.64 billion** for the half-year ending December 31, 2025.
- **OGDCL**: Experienced an 11% decline in cumulative profit-after-tax, dropping to **Rs 73 billion** for the first half of FY26.
- [Pakistan Oilfields Limited \(POL\)](#): Reported an H1 profit of **Rs 12.44 billion**, reflecting a 20.58% increase.  Profit by Pakistan Today +4

Revenue vs Profit

- In business finance, **revenue** is the total amount of money a company brings in from sales, while **profit** is the amount that remains after all expenses are paid.

Feature 	Revenue (Top Line)	Profit (Bottom Line)
Definition	Total income from business activities.	Income remaining after all costs are deducted.
Formula	$\text{Sales Price} \times \text{Quantity Sold.}$	$\text{Revenue} - \text{Total Expenses.}$
Statement Position	Appears at the very top of the income statement.	Appears at the very bottom of the income statement.
Indicates	Market demand and business scale.	Operational efficiency and sustainability.
Focus	Growth and market expansion.	Cost control and pricing optimization.

The Journey from Revenue to Profit

Companies track different "levels" of profit as they subtract various expenses from their revenue:  HiBob

1. **Gross Profit:** Revenue minus direct production costs (**COGS**).
2. **Operating Profit:** Gross profit minus daily operating expenses like rent, salaries, and marketing.
3. **Net Profit:** The final amount after subtracting taxes, interest, and any other remaining costs.  HiBob +1

Why the distinction matters

- **High Revenue ≠ Success:** A business can have millions in revenue but still lose money if its expenses (like massive advertising or high manufacturing costs) exceed that income.
- **Growth vs. Efficiency:** Startups often prioritize **revenue growth** to capture market share, while mature companies focus on **profitability** to reward shareholders and ensure long-term survival.
- **Decision Making:** Revenue helps determine manufacturing and inventory needs; profit dictates how much capital can be reinvested or paid out as dividends.  Investopedia +5

Financial Image

- **Financial Image** refers to the **overall perception or reputation of an individual, company, or organization regarding its financial strength, stability, and credibility**. It shows how others (investors, customers, banks, and the public) view the financial condition and reliability of that entity.

Key Aspects of Financial Image

- **Financial Stability**
 - The ability of a company or individual to manage expenses, debts, and income effectively.
- **Profitability**
 - How well a business generates profit from its operations.
- **Creditworthiness**
 - The level of trust banks, investors, and suppliers have in lending money or doing business.
- **Financial Transparency**
 - Clear and honest financial reporting that builds trust among stakeholders.
- **Reputation with Investors and Customers**
 - A strong financial image attracts investors and increases customer confidence.

Main statistical measures used to evaluate a company's financial image

1. Revenue Growth

- Shows how much a company's sales increase over time.
- Higher growth means the company is expanding and performing well.

Formula:

$$\text{Revenue Growth Rate} = \frac{\text{Current Revenue} - \text{Previous Revenue}}{\text{Previous Revenue}} \times 100$$

2. Profit Margin

- Indicates how much profit a company makes from its revenue.
- A higher profit margin means the company is managing its costs efficiently.

Formula:

$$\text{Profit Margin} = \frac{\text{Net Profit}}{\text{Revenue}} \times 100$$

Main statistical measures used to evaluate a company's financial image

3. Return on Investment (ROI)

- Measures how effectively a company uses investments to generate profit.

Formula:

$$\text{ROI} = \frac{\text{Net Profit}}{\text{Total Investment}} \times 100$$

4. Debt-to-Equity Ratio

- Shows how much debt the company uses compared to its own capital.
- Lower ratios generally indicate better financial stability.

Formula:

$$\text{Debt-to-Equity Ratio} = \frac{\text{Total Debt}}{\text{Shareholders' Equity}}$$

Main statistical measures used to evaluate a company's financial image

5. Current Ratio

- Measures the company's ability to pay short-term liabilities.

Formula:

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

6. Earnings Per Share (EPS)

- Indicates how much profit is earned for each share of stock.
- Investors use it to judge company profitability.

Formula:

$$EPS = \frac{\text{Net Profit}}{\text{Number of Outstanding Shares}}$$

Example

- If a software house consistently earns profit, pays its employees on time, and maintains transparent financial records, it develops a **strong financial image**. This helps the company attract investors and gain trust in the market.
- **Importance**
- Attracts **investors and funding**
- Builds **trust and credibility**
- Improves **business partnerships**
- Supports **long-term growth**
-  **In short:** Financial image is the **public perception of how financially strong and trustworthy a person or organization is.**

Well-known companies with a Strong Financial Image

1. Apple Inc.

- One of the most valuable companies in the world.
- Generates huge revenue from products like iPhone, Mac, and services.
- Has very high profits and strong cash reserves.

2. Microsoft

- A global leader in software, cloud computing, and AI.
- Known for strong profitability and a stable financial position.

3. Alphabet Inc. (parent company of Google)

- Generates large profits from online advertising, YouTube, and cloud services.
- Among the most profitable companies worldwide.

4. Amazon

- One of the largest e-commerce and cloud computing companies.
- Strong revenue growth and large global market share through AWS and online retail.

5. NVIDIA

- Leader in AI chips and high-performance computing.
- Rapid growth and very high profit margins due to demand for AI technology.

6. Visa

- One of the largest global payment processing companies.
- High profit margins and a strong global transaction network.

Software/Technology companies with a Strong Financial Image

1. Microsoft

- One of the largest software companies in the world.
- Major products include Windows, Office, and Azure cloud services.
- Strong financial position due to consistent profits and global demand.

2. Oracle Corporation

- Famous for database software and enterprise solutions.
- Provides cloud services and business management software.
- Known for stable revenue from corporate clients.

3. Adobe Inc.

- Develops popular creative software like Photoshop and Illustrator.
- Subscription model (Adobe Creative Cloud) generates steady income.
- Highly profitable in the digital media industry.

4. Salesforce

- Global leader in Customer Relationship Management (CRM) software.
- Provides cloud-based business solutions for companies worldwide.
- Strong growth and investor confidence.

5. SAP

- One of the largest enterprise software companies.
- Provides ERP and business management software used by large organizations.
- Very strong financial reputation in the corporate software market.

Software houses in Pakistan that are known for having a strong Financial Image

1. Systems Limited

- One of the **largest IT companies in Pakistan**.
- Provides digital transformation, cloud, and software development services.
- Listed on the Pakistan Stock Exchange and has strong financial performance.

2. NetSol Technologies

- A well-known Pakistani software company working in **global financial and leasing software solutions**.
- Has offices in many countries including the USA, UK, and China.
- Strong international clients and stable revenue.

3. 10Pearls

- A global digital technology company founded in Pakistan.
- Works in AI, mobile apps, and software development.
- Known for working with international clients and startups.

4. Arbisoft

- A well-reputed software house providing **software engineering and product development services**.
- Works with global companies and has a strong reputation in the tech industry.

5. Contour Software

- A subsidiary of a global company and one of the **largest IT employers in Pakistan**.
- Provides enterprise software development services.
- Financially strong due to international partnerships.

Companies in Pakistan that are known for having a strong financial image

- **1. Oil and Gas Development Company Limited (OGDCL)**
- One of the **most valuable companies in Pakistan**.
- Market value around **\$4.3 billion**, making it the top company by valuation.
- Plays a major role in Pakistan's oil and gas sector.

- **2. Habib Bank Limited (HBL)**
- One of the largest banks in Pakistan.
- Reported **revenue of about PKR 315.53 billion in 2024**.
- Strong profitability and large customer base.

- **3. Meezan Bank**
- Pakistan's largest Islamic bank.
- Market capitalization around **\$3.1 billion**.
- Known for strong growth and profitability.

Companies in Pakistan that are known for having a strong financial image

- **4. Lucky Cement**
 - One of the **largest cement producers in Pakistan** and part of the KSE-100 index.
 - Market value around **\$2.5 billion**.
 - Diversified business group with multiple subsidiaries.
- **5. Pakistan State Oil (PSO)**
 - Pakistan's largest oil marketing company.
 - Reported **revenue of about PKR 3.74 trillion in 2024** and profit of about **PKR 18.3 billion**.
- **6. State Life Insurance Corporation**
 - One of the largest insurance companies in Pakistan.
 - Holds the **highest financial strength rating (AAA)** and reported profit of about **PKR 18.3 billion in 2024**.

Cost benefit Evaluation Techniques

- Net Profit
- Payback period
- Return on investment (ROI)
- Net present value

1) Net Profit

$NP = \text{total income} - \text{total cost}$

- Very simple technique
- Does not consider time element
- Of limited use when used in isolation

Net Profit: Example

- In the example below, we have data in each column for each of the project. The project duration is 5 years. In order to calculate the Net Profit for a particular project, we need to add all the data in that column together.

- Net Profit

Year	Project 1	Project 2	Project 3	Project 4
0	-100,000	-1,000,000	-100,000	-120,000
1	10,000	200,000	30,000	30,000
2	10,000	200,000	30,000	30,000
3	10,000	200,000	30,000	30,000
4	20,000	200,000	30,000	30,000
5	100,000	300,000	30,000	75,000
Net Profit	50,000	100,000	50,000	75,000

2) Payback Period

- Payback period means “How soon can we get our cash back?” or break even
- Project with short payback periods are preferred nowadays.
- Does not consider income or expenditure after break even point is reached.
- If cash inflows are constant, use formula:

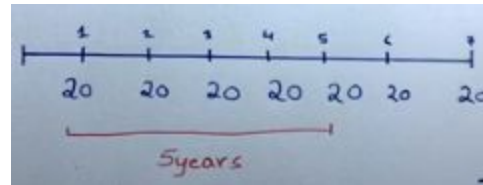
Initial investment

Annual cash inflow

- else use cumulative method

Payback Period: Example 1

- Investment = \$100



Initial investment = 100 = 5 years

Annual cash inflow 20

Payback Period: Example 2

- Initial investment = \$25,00,000
- Annual cash inflow = \$5,00,000 for 8 years
- $PBP = 2500000 / 500000 = 5 \text{ years}$

Payback Period: Example 3

- Calculate Payback Period for Machines A & B

	<u>Machine A</u>	<u>Machine B</u>
Initial investment	1500000	2000000
Cash inflow	500000	550000
	$1500000/500000$	$2000000/550000$
	3 years	3.6years

Answer: Go for Machine A because Machine A's Payback Period is shorter than Machine B's Payback Period.

Payback Period: Cumulative Method

When to use cumulative method for calculation of Payback Period?

Use cumulative method to calculate payback period when cash inflow is not constant.

- Using the **cumulative method**, you calculate the payback period by summing up the cash inflows year by year until the total equals or exceeds the initial investment.

Here are the steps to calculate the payback period using the cumulative method:

1. List the Initial Investment and Annual Cash Inflows

- Initial Investment (Cost):** The total amount spent on the project or investment.
- Annual Cash Inflows:** The expected or actual cash inflows from the investment over each period (e.g., yearly).

2. Calculate Cumulative Cash Flows

- Start with the initial cash inflow and keep adding the subsequent cash inflows to get the cumulative cash flows for each year.

3. Identify the Year When the Cumulative Cash Flow Equals or Exceeds the Initial Investment

- The payback period is the time at which the cumulative cash flows become equal to or exceed the initial investment.
- If it happens in between two years, you will calculate the fraction of the year required to reach the initial investment.

4. Calculate the Fraction (if necessary)

- If the investment is recovered between two years, calculate the fraction of the year as:

$$\text{Fraction of Year} = \frac{\text{Amount Remaining to Recover}}{\text{Cash Inflow in the Year}}$$

Formula Representation:

$$\text{Payback Period} = \text{Number of Full Years} + \frac{\text{Remaining Amount to Recover}}{\text{Cash Inflow in the Next Year}}$$

Payback Period: Cumulative Method

Example:

- Initial Investment: \$50,000
- Annual Cash Inflows:
 - Year 1: \$20,000
 - Year 2: \$15,000
 - Year 3: \$10,000
 - Year 4: \$10,000

Step 1: Calculate Cumulative Cash Flows

Year	Annual Cash Inflow	Cumulative Cash Flow
1	\$20,000	\$20,000
2	\$15,000	\$35,000
3	\$10,000	\$45,000
4	\$10,000	\$55,000

Step 2: Find the Year

The cumulative cash flow exceeds \$50,000 during Year 4. \$50,000 is initial investment

Step 3: Calculate the Fraction

$$\text{Remaining Amount to Recover} = 50,000 - 45,000 = 5,000$$

$$\text{Fraction of Year} = \frac{5,000}{10,000} = 0.5 \text{ years}$$

5000 is the amount above the initial investment in the 4th year, that is \$55,000/10,000 is the annual cash inflow if the 4th year

Step 4: Final Payback Period

$$\text{Payback Period} = 3 + 0.5 = 3.5 \text{ years}$$

In the 4th year, the cumulative cash inflow exceeds the initial investment so it means it is 3 years + fraction of the year (calculated above)